Post-Test Assessment

Study: Evaluating AI Guardrails and Their Effect on Students' Independent Critical Thinking

Instructions

- This is a post-test assessment to evaluate your independent understanding.
- Do NOT use any AI tools (e.g., ChatGPT, Gemini).
- Answer all questions to the best of your ability.
- Provide clear reasoning and explanation for each answer.
- Total Questions: 5
- Time: 25–30 minutes

Questions

Q1. Overfitting and Model Performance

A classification model achieves 95% accuracy on training data but only 60% accuracy on test data.

a) What is happening in this scenario?
b) Suggest two ways to improve the model and explain why they would help.

Q2. Evaluation Metrics

You are given a confusion matrix for a binary classifier.

a) Explain what precision and recall represent.
b) Describe a real-world scenario where recall is more important than precision.

Q3. Model Selection

You are solving a classification problem and have the option to use either:

- A Decision Tree
- A Neural Network

When would you prefer a **Decision Tree over a Neural Network**?
Provide reasoning based on interpretability, data size, or complexity.

Q4. Bias vs Variance

Explain the difference between:

- **High bias**
- **High variance**

Provide one example of each and suggest one method to address them.

Q5. Feature Engineering and Model Improvement

You are given a dataset with several features, but the model performance is poor.

a) What steps would you take to improve the model?
b) Explain the role of **feature engineering** in improving performance.